

Toy Object Parameters Guide

Plain-language reference for the Interactive Object Recovery tool

Source: `Foreground Masking/interactive object recovery.py`

Short answer: The Toy object model now has three available profiles: Gaussian star, Star cluster, and Compact galaxy. Location, brightness, FWHM, and truth dilation apply broadly. Axis ratio and object PA apply only to Compact galaxy. Toy objects must sit wholly inside the same deprojected, bar-aligned galaxy investigation area used by the normal image/profile reports.

Object Types

Type	Profile used	Best interpreted as	Important notes
Gaussian star	Single circular 2D Gaussian profile.	A compact point-source test object with a smooth, symmetric core.	Characterised by centroid, brightness, and FWHM. Uses FWHM to set Gaussian sigma. Ignores axis ratio and object PA.
Star cluster	Blend of three offset circular Gaussian components.	A small unresolved or barely resolved clump rather than a single isolated star.	Characterised by one centroid, brightness, and FWHM. FWHM sets the scale for all component offsets and widths. Ignores axis ratio and object PA.
Compact galaxy	Single elliptical Gaussian profile.	A small galaxy-like contaminant with an elongated shape.	Characterised by centroid, brightness, major-axis FWHM, axis ratio, and object PA.

Parameter Glossary

Parameter	Meaning	Applies to	Practical note
Type	Chooses the mathematical profile used to generate the injected Toy object.	All types	The internal values are gaussian, cluster, and galaxy.
Brightness	Chooses how the object brightness is set: either by peak residual sigma or by integrated magnitude.	All types	Peak residual sigma is usually the simpler test mode. Integrated magnitude is useful when the FITS photometric calibration is valid.
x deproj [arcsec], y deproj [arcsec]	Position of the Toy object in the deprojected, bar-aligned coordinate system.	All types	The code converts this deprojected position back to observed image pixels before injecting the model, then rejects placements outside the investigated bar-aligned cutout.
peak [resid sigma]	Peak brightness expressed as a multiple of the robust residual-image noise.	All types, when Brightness is Peak residual sigma	Example: 30 means the model peak is 30 times the robust residual sigma.
integrated mag	Target total magnitude for the whole injected object, not just the central pixel.	All types, when Brightness is Integrated magnitude	The code scales a unit-peak model so its integrated flux matches this magnitude.
zero flux [Jy]	Flux density of a zero-magnitude source, in Jansky.	Magnitude mode	The default 280.9 Jy is appropriate for Spitzer/IRAC 3.6 micron Vega magnitudes. Change it for another band or magnitude system.
FWHM [arcsec]	Full width at half maximum: the diameter across the profile where the brightness has fallen to half the peak.	All types	Converted to pixels using the galaxy pixel scale. For Gaussian profiles, $\text{sigma} = \text{FWHM} / 2.3548$.

Parameter	Meaning	Applies to	Practical note
axis ratio	Minor-axis width divided by major-axis width. A value of 1.0 is circular; smaller values are more elongated.	Compact galaxy only	In the current code, Gaussian star and Star cluster are circular and ignore this value.
object PA [deg]	Position angle of the object's major axis, measured in image-pixel coordinates.	Compact galaxy only	Only matters when the model is elliptical. It is not used for circular profiles.
truth dilation [px]	Extra pixel dilation applied to the truth mask after thresholding the injected model.	All types	This changes the evaluation mask used for recall/overlap, not the injected object's light profile.

Applicability Matrix

Parameter	Gaussian star	Star cluster	Compact galaxy
Type	Yes	Yes	Yes
Brightness mode	Yes	Yes	Yes
x/y deprojected location	Yes	Yes	Yes
peak residual sigma	Yes	Yes	Yes
integrated mag	Yes	Yes	Yes
zero flux [Jy]	Magnitude mode	Magnitude mode	Magnitude mode
FWHM [arcsec]	Yes	Yes	Yes
axis ratio	No	No	Yes
object PA [deg]	No	No	Yes
truth dilation [px]	Yes	Yes	Yes

Key Terms and Formulas

Gaussian sigma and FWHM

A Gaussian profile is often described by sigma, but the UI asks for FWHM because it is easier to interpret visually. The code uses:

Formula: $\text{sigma_pixels} = \max(0.2, \text{FWHM_pixels} / 2.3548)$

A larger FWHM makes the object broader. For the Star cluster profile, FWHM also sets how far apart the three Gaussian components are placed.

How the three object types are characterised

Each Toy object type is deliberately simple. The aim is to test recovery behaviour with controlled source shapes, not to fit a complete astrophysical model.

- Gaussian star: a single circular Gaussian. It is described by its deprojected position, brightness, and FWHM. It has no elongation or orientation.
- Star cluster: a fixed three-Gaussian blend. It is described by one central position, one brightness scale, and one FWHM scale; the component offsets and relative intensities are fixed by the tool.

- Compact galaxy: a single elliptical Gaussian. It is described by position, brightness, major-axis FWHM, axis ratio, and position angle.

Investigation-area constraint

Toy objects are only valid inside the galaxy area actually being investigated. In practice this is the same deprojected, bar-aligned square cutout used by the interactive displays, PNG reports, isophote views, and bar-major profiles. The horizontal axis is the deprojected bar-major axis and the vertical axis is the deprojected bar-minor axis.

- The toy centre must fall inside this cutout.
- The full truth mask, after the 8 percent model threshold and optional truth dilation, must also remain inside the cutout.
- Toy-object optimisation results produced before this constraint was added should be treated as superseded, because those older runs could place toys anywhere in the finite FITS footprint.

Integrated magnitude mode

Magnitude mode first builds the chosen object profile with peak = 1, then scales that unit model so the total integrated flux matches the requested magnitude.

Formula: $\text{flux_Jy} = \text{zero_flux_Jy} * 10^{(-0.4 * \text{magnitude})}$

If the FITS header uses BUNIT=MJy/sr, the code converts summed model intensity to Jy using the pixel area in steradians. If not, it looks for a count-based magnitude zero point in MAGZP, MAGZERO, or ZEROPOINT.

Truth mask and recovery metrics

The truth mask is not the same as the light profile. The code thresholds the model at 8 percent of the peak or 8 percent of the maximum model value, then optionally dilates that binary mask by truth dilation [px].

- Recall is recovered truth pixels divided by total truth pixels.
- Incremental recall uses only the new mask pixels produced after injecting the Toy object.
- Incremental precision asks what fraction of the new mask pixels overlap the truth mask.

Suggested Usage

- Use Gaussian star for a simple compact-source sensitivity test.
- Use Star cluster when you want a blended compact contaminant rather than a single source.

- Use Compact galaxy when elongation and orientation matter; this is the only Toy object type that uses axis ratio and object PA.
- Use Peak residual sigma for quick recovery experiments. Use Integrated magnitude when you want physically calibrated brightness and the FITS header supports the conversion.
- Place Toy objects inside the displayed bar-aligned investigation area; if the model or truth mask spills outside that area, the tool rejects the placement.
- Remember that truth dilation changes the scoring mask, not the injected object itself.

Systematic documentation review - 16 August 2026

Review classification: Reviewed - Toy Object construction reference remains valid

Updates made: Object morphology, placement, truth-mask and recovery definitions were reviewed. A dated link to the final optimisation outcome was added.

Current interpretation: The guide defines injected objects and metrics; it does not prescribe the final SEP or MTOjects production parameter set.

Authoritative combined outcome: SEP and MTOjects Toy Objects Optimisation Summary and Conclusions.docx. Full-precision results: Foreground Masking Optimisation Results.xlsx.